

Experiment (10)

Border Gateway Protocol (BGP)

10.1 Objectives

To achieve importance of BGP

Border Gateway Protocol (BGP)

10.2 Equipment / Program Requirements:

Ensp

Cisco packet tracker

10.3 Experiment Setup:

Make topology

Then put ips for each device

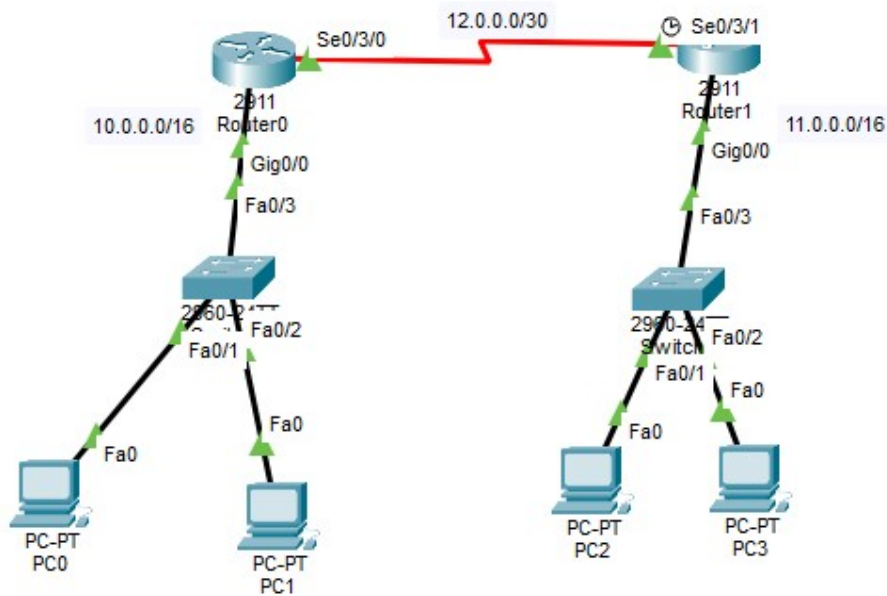
Then configuration BGP Protocol Interior

Then Ping For Troubleshooting

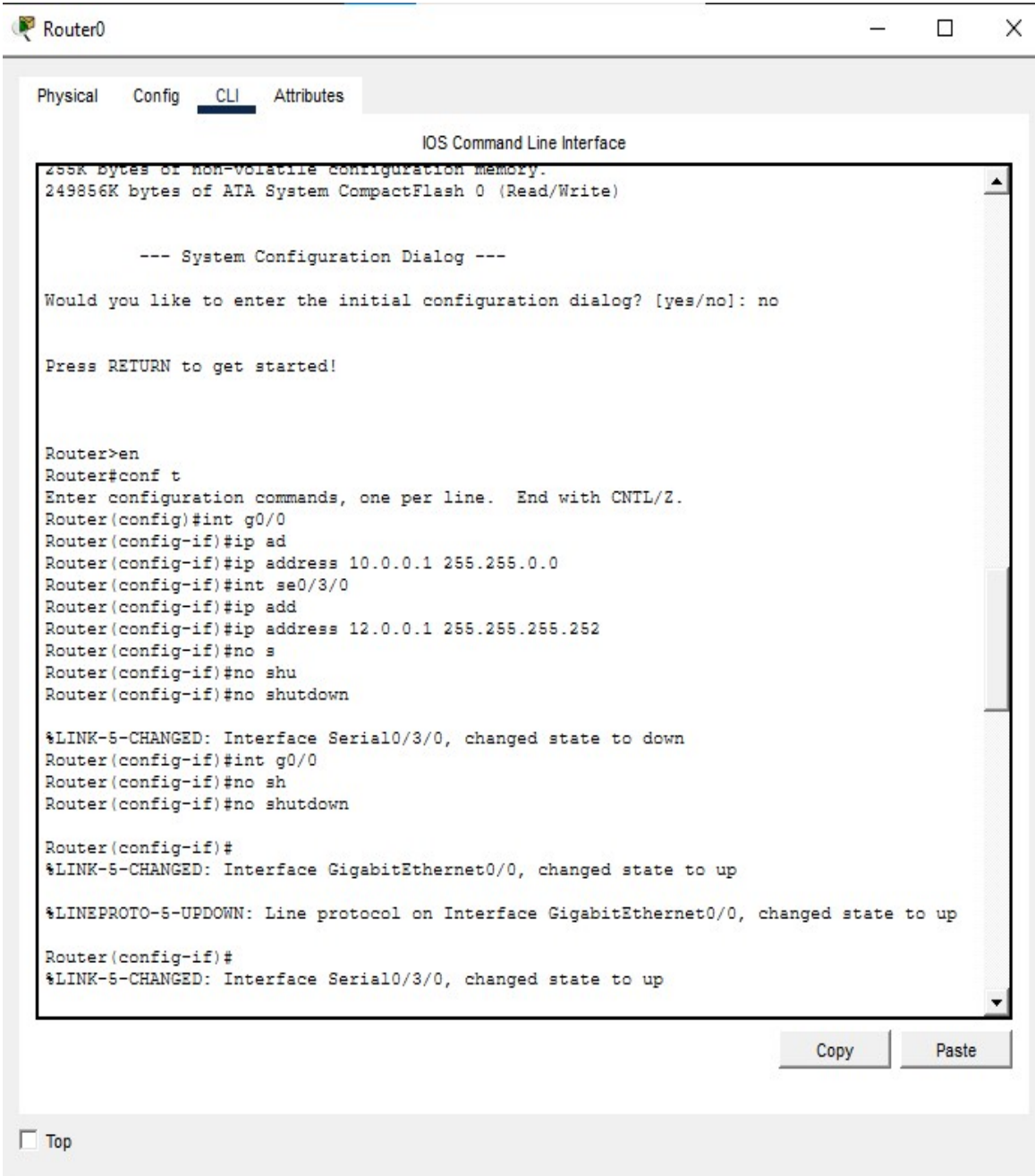
10.4 Simulation Results:

Cisco Packet Traker:

Topology



Router 0 Configurtaion:



Router 0 Routing Table :

Router0

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/30 is directly connected, Serial0/3/0
L    12.0.0.1/32 is directly connected, Serial0/3/0

Router(config)#%BGP-5-ADJCHANGE: neighbor 12.0.0.2 Up

Router(config)#do sh
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/16 is directly connected, GigabitEthernet0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
    11.0.0.0/16 is subnetted, 1 subnets
B    11.0.0.0/16 [20/0] via 12.0.0.2, 00:00:00
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/30 is directly connected, Serial0/3/0
L    12.0.0.1/32 is directly connected, Serial0/3/0

Router(config)#do sh ip bgp
BGP table version is 3, local router ID is 12.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 10.0.0.0/16     0.0.0.0               0      0 32768 i
*> 11.0.0.0/16     12.0.0.2               0      0      0 200 i

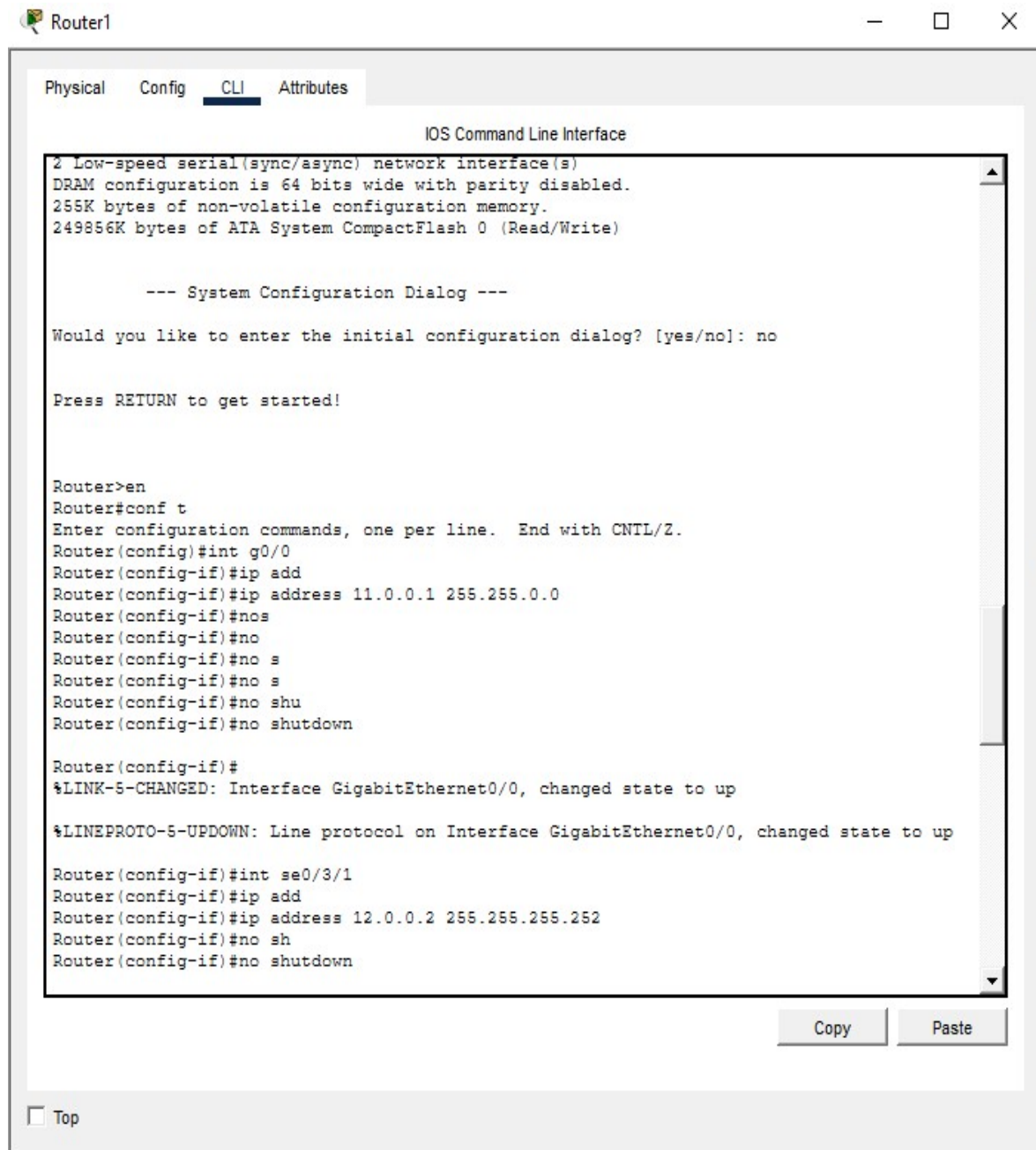
Router(config)#

```

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Router 1 Configuration:



Router 1 Routing Table:

Router1

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Router(config-router)#neighbor 12.0.0.1 r
Router(config-router)#neighbor 12.0.0.1 remote-as 100
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 12.0.0.1 Up

Router(config-router)#network 11.0.0.0 mask 255.255.0.0
Router(config-router)#exit
Router(config)#do sh
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/16 is subnetted, 1 subnets
B       10.0.0.0/16 [20/0] via 12.0.0.1, 00:00:00
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/16 is directly connected, GigabitEthernet0/0
L       11.0.0.1/32 is directly connected, GigabitEthernet0/0
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, Serial0/3/1
L       12.0.0.2/32 is directly connected, Serial0/3/1

Router(config)#do sh
Router(config)#do sh ip bgp
BGP table version is 3, local router ID is 12.0.0.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               x RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
*> 10.0.0.0/16      12.0.0.1             0      0      0 100 i
*> 11.0.0.0/16      0.0.0.0              0      0 32768 i

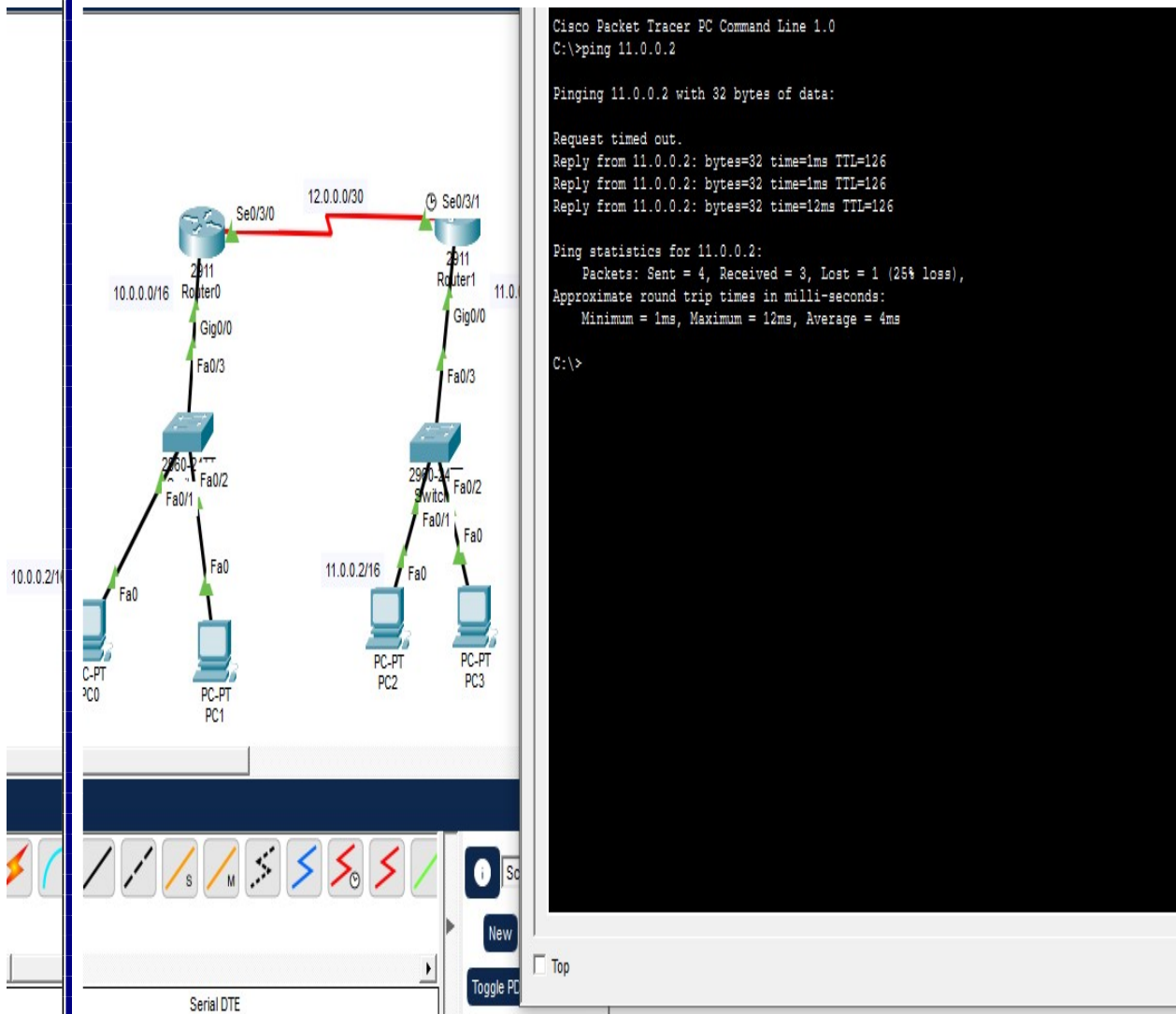
Router(config)#

```

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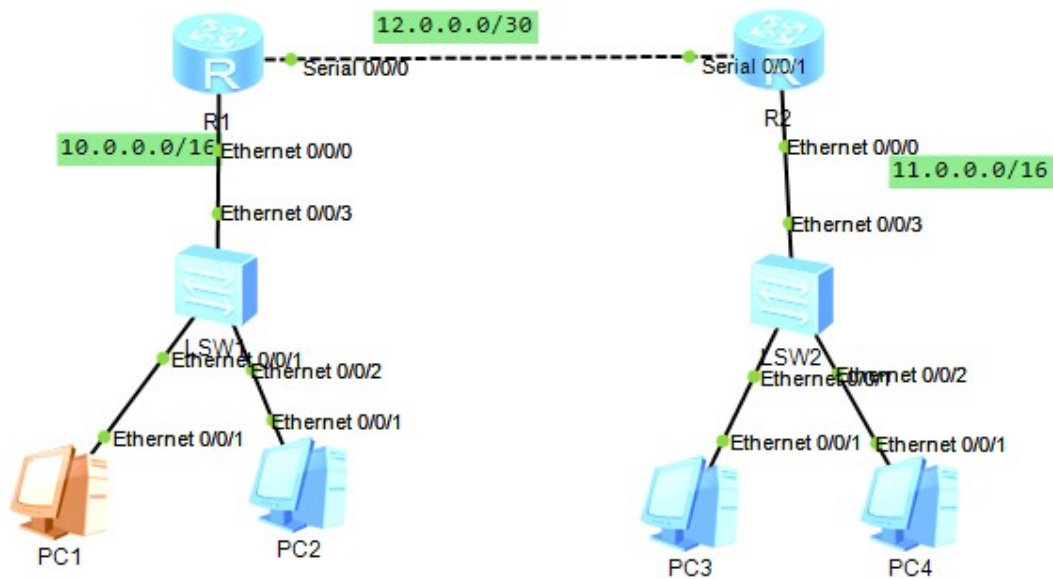
☐ Top

Testing:



eNSP:

Topology:



R1 Configuration:

R1

Apr 23 2026 14:56:38-08:00 Huawei %%01IFNET/4/LINK_STATE(1)[2]:The line protocol PPP on the interface Serial0/0/0 has entered the UP state.

<Huawei>

<Huawei>u t m

Info: Current terminal monitor is off.

<Huawei>u t l

Info: Current terminal logging is off.

<Huawei>sys

Enter system view, return user view with Ctrl+Z.

[Huawei]int eth0/0/0

[Huawei-Ethernet0/0/0]ip add

[Huawei-Ethernet0/0/0]ip address 10.0.0.1 16

[Huawei-Ethernet0/0/0]int se0/0/0

[Huawei-Serial0/0/0]ip ad

[Huawei-Serial0/0/0]ip address 12.0.0.1 30

[Huawei-Serial0/0/0]q

[Huawei]bgp 100

[Huawei-bgp]peer 12.0.0.2 a

[Huawei-bgp]peer 12.0.0.2 as-path-filter

[Huawei-bgp]peer 12.0.0.2 advertise-community

[Huawei-bgp]peer 12.0.0.2 advertise-ext-community

[Huawei-bgp]peer 12.0.0.2 as-number 200

[Huawei-bgp]q

[Huawei]dis

[Huawei]display io

[Huawei]display ip

[Huawei]display ip-subnet-vlan

[Huawei]display ipv6

[Huawei]display ip

[Huawei]display ipsec

[Huawei]display ip-subnet-vlan

[Huawei]display ipv6

[Huawei]display ip

[Huawei]display ipsec

[Huawei]display ip-ro

[Huawei]display ip-ro

[Huawei]display ip rou

[Huawei]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 7

Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
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R1 Routing Table:

R1

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/16	Direct	0	0	D	10.0.0.1	Ethernet0/0/0
10.0.0.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
11.0.0.0/16	EBGP	255	0	D	12.0.0.2	Serial0/0/0
12.0.0.0/30	Direct	0	0	D	12.0.0.1	Serial0/0/0
12.0.0.1/32	Direct	0	0	D	127.0.0.1	Serial0/0/0
12.0.0.2/32	Direct	0	0	D	12.0.0.2	Serial0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei]bgp 100

[Huawei-bgp]network 10.0.0.0 16

[Huawei-bgp]q

[Huawei]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 8 Routes : 8

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/16	Direct	0	0	D	10.0.0.1	Ethernet0/0/0
10.0.0.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
11.0.0.0/16	EBGP	255	0	D	12.0.0.2	Serial0/0/0
12.0.0.0/30	Direct	0	0	D	12.0.0.1	Serial0/0/0
12.0.0.1/32	Direct	0	0	D	127.0.0.1	Serial0/0/0
12.0.0.2/32	Direct	0	0	D	12.0.0.2	Serial0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei]display bgp peer

BGP local router ID : 10.0.0.1

Local AS number : 100

Total number of peers : 1 Peers in established state : 1

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	Pre
12.0.0.2	4	200	4	5	0	00:01:48	Established	
1								

[Huawei]

R2 Configuration:

```
R2
The device is running!

<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]int eth0/0/0
[Huawei-Ethernet0/0/0]ip ad
[Huawei-Ethernet0/0/0]ip address 11.0.0.1 16
[Huawei-Ethernet0/0/0]int se0/0/1
[Huawei-Serial0/0/1]ip ad
[Huawei-Serial0/0/1]ip address 12.0.0.2 30
[Huawei-Serial0/0/1]q
[Huawei]
[Huawei]bgp 200
[Huawei-bgp]peer 12.0.0.1 as
[Huawei-bgp]peer 12.0.0.1 as-
[Huawei-bgp]peer 12.0.0.1 as-path-filter
[Huawei-bgp]peer 12.0.0.1 as-number
[Huawei-bgp]peer 12.0.0.1 as-n
[Huawei-bgp]peer 12.0.0.1 as-number 200
[Huawei-bgp]peer 12.0.0.1 as-number 100
Error: The peer already exists in AS 200.
[Huawei-bgp]peer 12.0.0.1 as-number 200
Error: The peer already exists in AS 200.
[Huawei-bgp]undo peer 1.0.0
      ^
Error:Incomplete command found at '^' position.
[Huawei-bgp]
[Huawei-bgp]undo peer 12.0.0.1
[Huawei-bgp]peer 12.0.0.1 as-number 100
[Huawei-bgp]network 11.0.0.0 16
[Huawei-bgp]q
[Huawei]display ip ro
[Huawei]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
```

R2 Routing Table:

```

R2
[Huawei]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

Destination/Mask    Proto    Pre  Cost    Flags NextHop        Interface
-----
11.0.0.0/16 Direct  0    0        D  11.0.0.1    Ethernet0/0/0
11.0.0.1/32 Direct  0    0        D  127.0.0.1    Ethernet0/0/0
12.0.0.0/30 Direct  0    0        D  12.0.0.2    Serial10/0/1
12.0.0.1/32 Direct  0    0        D  12.0.0.1    Serial10/0/1
12.0.0.2/32 Direct  0    0        D  127.0.0.1    Serial10/0/1
127.0.0.0/8 Direct  0    0        D  127.0.0.1    InLoopBack0
127.0.0.1/32 Direct  0    0        D  127.0.0.1    InLoopBack0

[Huawei]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

Destination/Mask    Proto    Pre  Cost    Flags NextHop        Interface
-----
11.0.0.0/16 Direct  0    0        D  11.0.0.1    Ethernet0/0/0
11.0.0.1/32 Direct  0    0        D  127.0.0.1    Ethernet0/0/0
12.0.0.0/30 Direct  0    0        D  12.0.0.2    Serial10/0/1
12.0.0.1/32 Direct  0    0        D  12.0.0.1    Serial10/0/1
12.0.0.2/32 Direct  0    0        D  127.0.0.1    Serial10/0/1
127.0.0.0/8 Direct  0    0        D  127.0.0.1    InLoopBack0
127.0.0.1/32 Direct  0    0        D  127.0.0.1    InLoopBack0

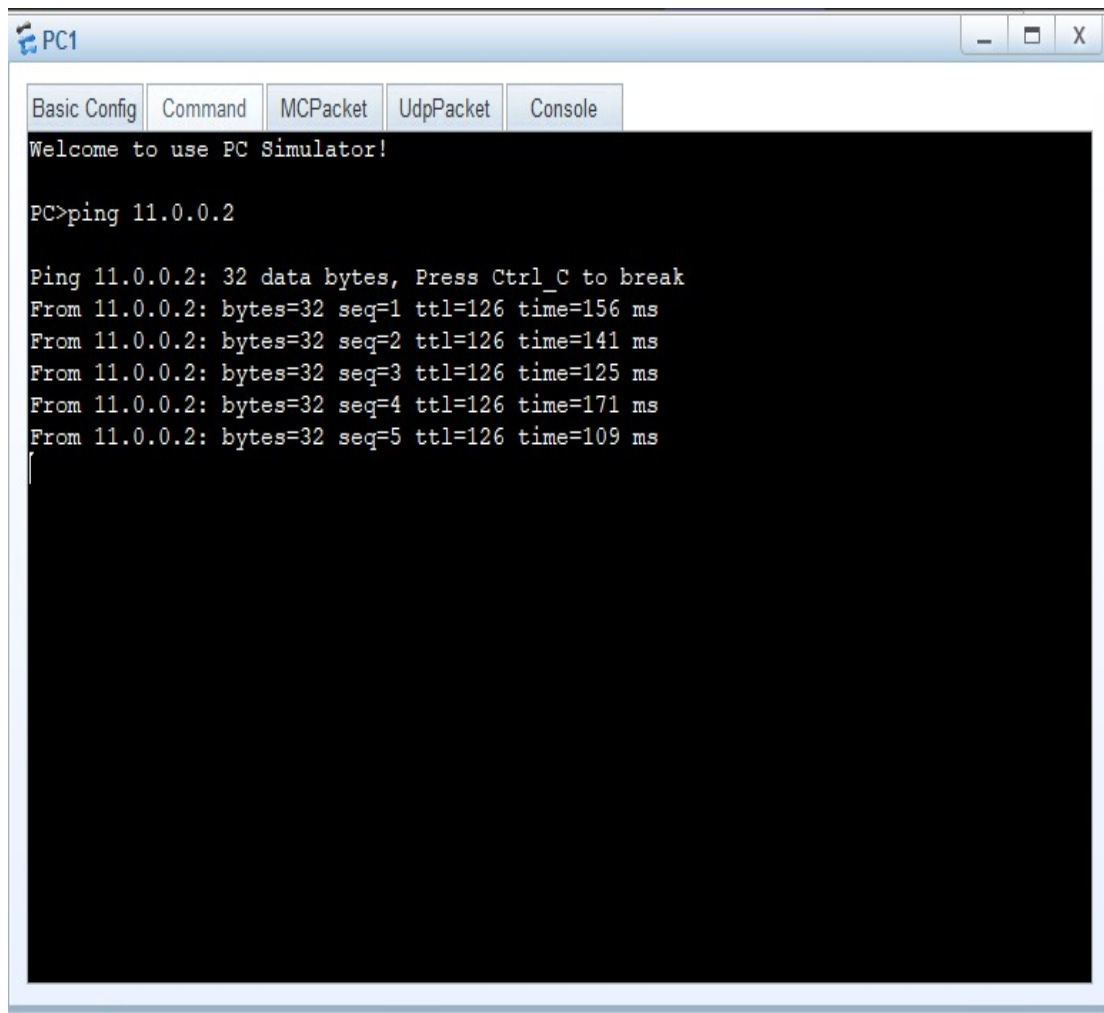
[Huawei]display bgp peer

BGP local router ID : 11.0.0.1
Local AS number : 200
Total number of peers : 1          Peers in established state : 1

Peer          V          AS  MsgRcvd  MsgSent  OutQ  Up/Down      State Pre
fRcv
-----
12.0.0.1      4          100    4        4        0 00:01:58 Established
1
[Huawei]

```

Testing:



10.5 Conclusions:

Border Gateway Protocol (BGP) is the core routing protocol of the Internet, responsible for connecting different Autonomous Systems (AS).

It uses a **path vector mechanism** to select the best route based on attributes such as **AS-Path** and **Next-Hop**, rather than just the shortest path.

BGP is highly **scalable** and supports **policy-based routing**, which allows network administrators to control how traffic flows between networks.

However, it is more **complex to configure** compared to other routing protocols like OSPF or RIP.